

AI/ML intern DAY 11

TASK — 3D Virtual Try-On Research

LOYA PRANAY (pranay8679@gmail.com)

Date: 25-05-2026

Introduction

The objective of this task was to research AI-powered 3D Virtual Try-On Systems used in modern digital fashion platforms and metaverse applications.

The research focused on:

- 3D avatar generation
- body mesh reconstruction
- cloth simulation
- motion tracking
- real-time fitting systems

The task explored how Artificial Intelligence and Computer Vision technologies create realistic virtual fitting experiences using digital human models and cloth simulation workflows.

Research and Exploration

3D Avatars

Studied AI-based avatar generation systems used for:

- digital humans
- metaverse applications
- gaming environments
- virtual shopping systems

Features Explored

- avatar rendering
- digital body modelling
- facial synchronization
- movement tracking
- avatar customization

The research helped understand how AI creates realistic human avatars for virtual try-on applications.

Body Mesh Generation

Researched body mesh generation workflows for reconstructing human body structure in 3D environments.

Technologies Explored

- SMPL body models
- body reconstruction systems
- AI body scanning
- mesh generation frameworks

Features Studied

- body structure generation
- skeletal mesh modelling
- body shape reconstruction
- realistic body rendering

The body mesh system converts body landmarks into digital 3D body structures.

Cloth Simulation

Studied cloth simulation systems for realistic garment movement and fitting.

Features Researched

- fabric movement
- dynamic cloth behaviour
- body-cloth interaction
- garment deformation
- realistic cloth rendering

Applications

- fashion technology
- gaming systems
- metaverse clothing
- virtual fitting rooms

The simulation workflow improves realism in virtual clothing systems.

Real-Time Fitting Systems

Explored AI-based real-time fitting systems for live virtual try-on applications.

Features Studied

- live cloth fitting
- avatar synchronization
- motion-based garment fitting
- real-time rendering

Benefits

- instant virtual fitting
- personalized fashion analysis
- interactive shopping experience

The research focused on improving real-time responsiveness in digital fitting systems.

Motion Tracking

Researched motion tracking technologies used for:

- avatar movement
- gesture tracking
- body synchronization
- pose analysis

Features Explored

- real-time body tracking
- gesture recognition
- movement synchronization
- avatar motion mapping

Motion tracking helps synchronize digital clothing with user body movement.

Tools Explored

Tool	Purpose
Unity	3D rendering and simulation
Blender	Cloth simulation and avatar modelling
Three.js	Web-based 3D rendering
SMPL Models	Human body mesh generation
AR Frameworks	Real-time augmented reality fitting

Research Topics Covered

- 3D body reconstruction
- realistic cloth fitting
- avatar rendering
- real-time try-on systems
- cloth simulation workflows
- digital human generation

Basic Blender Cloth Simulation Workflow

Step 1 — Create 3D Avatar

Generate:

- body mesh
- skeletal structure
- avatar model

Step 2 — Import Clothing Model

Import:

- shirt
- hoodie
- pants
- garments

into Blender.

Step 3 — Apply Cloth Physics

Enable:

- cloth simulation
 - collision detection
 - fabric physics
-

Step 4 — Run Simulation

The cloth dynamically:

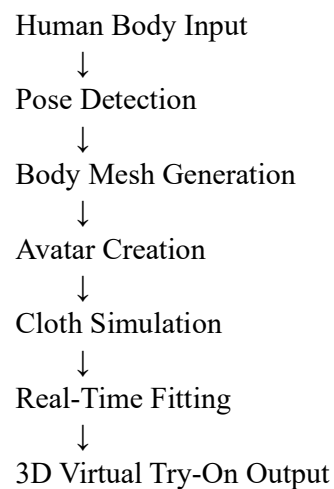
- bends
 - stretches
 - fits avatar body
-

Step 5 — Render Final Output

Generate:

- virtual fitting output
 - realistic garment rendering
 - avatar preview
-

Basic 3D Workflow Architecture



Technologies Used

- Blender

- Unity
 - Three.js
 - SMPL Models
 - Artificial Intelligence
 - Machine Learning
 - Computer Vision
 - Deep Learning
-

Expected Outputs

- 3D avatar generation
 - body mesh visualization
 - cloth simulation workflows
 - realistic garment fitting
 - virtual fitting system research
 - AI rendering workflows
-

Research Findings

The research helped understand:

- realistic garment simulation
- body mesh reconstruction
- AI avatar rendering
- real-time fitting systems
- motion synchronization workflows

The study demonstrated how AI-powered 3D Virtual Try-On Systems generate realistic digital fitting experiences using Computer Vision and Deep Learning technologies.

Learning Outcomes

During this task, the following concepts and skills were learned:

- 3D avatar generation
- body mesh reconstruction
- cloth simulation systems

- motion tracking technologies
 - avatar synchronization
 - real-time fitting workflows
 - digital fashion technologies
 - AI rendering systems
-

Conclusion

The 3D Virtual Try-On Research task successfully explored how Artificial Intelligence and Computer Vision technologies generate realistic digital fitting systems using avatars, cloth simulation, and body mesh reconstruction workflows.

The task improved understanding of:

- 3D avatar systems
- cloth simulation workflows
- body mesh generation
- motion tracking
- real-time fitting systems
- digital fashion technologies
- AI virtual try-on applications